

Tokenizer and Sequence-Length Choices for SLMs in Agglutinative Korean: A Fair Comparison of MeCab and SentencePiece

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Abstract

We study how **tokenizer choice** (SentencePiece BPE/Unigram vs. MeCab) and **sequence-length policy** (95th vs. 98th percentile) affect Korean sentiment classification (NSMC) under small/efficient language model (SLM) constraints where large LLMs are impractical. Using a clean baseline (tab-separated raw data with NA-row removal only), SentencePiece is trained with nmt_nfkc normalization, character_coverage=0.9995, and byte_fallback=0; MeCab-ko provides morphology-aware token sequences. With an identical RNN backbone and training setup, **MeCab (95p)** achieves the best **Accuracy of 0.8616**, followed by **SentencePiece-BPE (8k)** 0.8525, **SentencePiece-BPE (10k)** 0.8500, and **SentencePiece-Unigram (8k)** 0.8391. Expanding the cut-off from **95p** to **98p** reduces MeCab’s accuracy from **0.8616** to **0.8418**. Findings indicate that (i) **morphology-aware tokenization** benefits RNN-based SLMs for agglutinative Korean, and (ii) **length/padding policy is a first-order factor**. Given the observed prevalence of repeated characters, numerals, and symbols in SP pieces, we argue that repeat/symbol normalization, Unigram Subword Regularization, enabling byte_fallback, and vocabulary sweeps could further narrow the gap.

I Introduction

While Large Language Models (LLMs) set new performance ceilings, their training/serving costs and privacy/latency constraints often make **Small Language Models (SLMs)** a practical choice for **on-device or resource-limited** scenarios. Given the tighter capacity budget of SLMs, **tokenizer selection** and **sequence-length policies** exert a disproportionately large impact on downstream accuracy and efficiency.

For **agglutinative languages** such as **Korean**, affixation and morphological variation can render whitespace-based tokenization inefficient, inflating sequence lengths. In practice, two families

dominate: (a) **morphology-aware analyzers** (e.g., MeCab-ko) and (b) **subword models** (SentencePiece with **BPE** or **Unigram**). These families trade off interpretability, language independence, OOV handling, and deployment simplicity. Moreover, **max-length cutoffs (percentile-based)** influence padding ratios and long-range dependencies, becoming a decisive factor for **RNN-based SLMs**.

This instructional study presents a controlled comparison on Korean text classification, evaluating **MeCab** (morphology-based) against **SentencePiece** (BPE/Unigram with varying vocabulary sizes) under identical budgets with an **RNN encoder**. We further quantify the effect of **95th vs 98th percentile** max-length policies. In brief, (i) **MeCab (95p)** yields the best accuracy, (ii) extending **max_len** to **98p** degrades performance—consistent with increased padding and information dilution in RNNs, and (iii) subword variants may narrow the gap when combined with **aggressive normalization** of repetitive tokens (e.g., “ㅋㅋㅋㅋ”) and **Unigram’s subword regularization**. These findings suggest **morphology-aware tokenization** and a **conservative cutoff (~95p)** as sensible defaults for **Korean SLMs on device**.

Our contributions are:

1. A **fair, same-budget** comparison of **morphology vs subword** tokenization for Korean SLM classification.
2. An empirical demonstration that **sequence-length policy** (95p vs 98p) **materially affects** RNN performance.

II Related works

1. Subword tokenization: BPE, Unigram, and normalization/regularization

Subword tokenization has become a standard technique for alleviating the out-of-vocabulary (OOV) problem and improving data efficiency. **Byte-Pair Encoding (BPE)** (Sennrich et al., 2016) constructs a vocabulary by merging frequent character pairs, significantly enhancing rare-word handling in multilingual tasks such as translation. **SentencePiece** (Kudo & Richardson, 2018) is language-independent and can be trained directly on raw text, thereby simplifying preprocessing pipelines. SentencePiece supports both **BPE** and the **Unigram LM** model (Kudo, 2018; Subword Regularization); the latter introduces probabilistic variation in segmentation during training, which improves robustness under domain shift or low-resource conditions.

2. Morphology-aware tokenization for agglutinative languages (Korean)

In agglutinative languages such as Korean and Japanese, tokenization that respects **morpheme boundaries** shortens sequences and makes syntactic and semantic information more explicit. In Japanese, the MeCab family of analyzers has been widely used due to its high accuracy (capturing tense, politeness, case markers, etc.), and in Korean, variants such as MeCab-ko have become practical standards. More recently, studies have applied **deep learning re-ranking** to morphological

analysis in Korean to reduce errors. Additionally, approaches specialized for Korean—such as **sub-character (jamo) decomposition** or **hybrid strategies**—have been reported to yield meaningful performance improvements.

3. Sequence length, padding, and sensitivity of small/recurrent models

RNN -based models are sensitive to long sequences, where issues such as **long-range dependency loss** and **increased padding ratio** degrade performance. While Transformers have alleviated much of this problem, in **small/on-device models (SLMs)**, lightweight recurrent structures remain practical choices. In this study, the observed performance drop when increasing **max_len from 95% to 98%** can be explained by this typical pattern: more padding and diluted information density leading to accuracy degradation.

III Method

Dataset

- NSMC (Naver Sentiment Movie Corpus) with the original split: 150K train / 50K test.
- **Missing values** before cleaning: 5 (train document), 3 (test). → **Row-wise removal**; no NA afterward.
- **Duplicates**: none in train/test.
- **Length stats (characters, train)**: mean 35.20, median 27, std 29.53, max 146.

Preprocessing

- Loaded tab-separated files; aside from dropping NA rows, **no extra normalization** (to serve as a clean baseline).
- Length capping is applied **after tokenization** using percentile thresholds (“Length Policy”).

Tokenization

- **SentencePiece** (v0.2.1)
 - Algorithms: **BPE** and **Unigram**.
 - **Vocab**: 8,000, 10,000.
 - **Normalizer**: nmt_nfkc (dummy prefix, remove extra whitespaces, escape whitespaces).
 - **character_coverage**: 0.9995; **byte_fallback=0**; 16 threads.
 - Training input: ratings_train.txt (150,001 sentences).
- **MeCab-ko** (via KoNLPy 0.6.0)
 - Morphology-based token sequences.

Length policy (max_len)

- **95th** or **98th** percentile of token length per configuration.

- Sequences exceeding the cutoff are **right-truncated**; others are padded.

Model

- **RNN-based classifier** (PyTorch).
 - The same model/training routine is used across runs; only tokenizer and length policy vary

Training & Evaluation

- Frameworks: PyTorch 2.7.1+ cu118, NumPy 2.2.6, Matplotlib 3.10.3.
- Metric: **Accuracy (primary)**, given near-balanced labels.
- For each condition (tokenizer $\times$ vocab $\times$ max_len), train with identical settings and evaluate on the **test set**.

IV Result

Quantitative (test Accuracy)

- MeCab (max_len 95%): 0.8616 $\leftarrow$ Best
- SentencePiece (BPE, vocab 8k): 0.8525
- SentencePiece (Unigram, vocab 8k): 0.8391
- SentencePiece (BPE, vocab 10k): 0.8500
- MeCab (max_len 98%): 0.8418

Interpretation

1. **Advantage of morphology-aware tokenization**
 - With the same backbone/training, **MeCab(95p)** tops the chart. In agglutinative Korean, respecting morphological boundaries shortens sequences and raises information density—beneficial for RNNs.
2. **Length policy is first-order**
 - Expanding to **98p hurts accuracy (0.8616 $\rightarrow$ 0.8418)**. Given the long-tail length distribution, this is consistent with **higher average padding** and **RNN sensitivity to longer sequences**.
3. **Narrow gap vs. subword; room for SP gains**
 - SP-BPE (8k/10k) and SP-Unigram (8k) trail **MeCab(95p)** by a small margin. SP training logs surface many pieces for **repeated laughter** (e.g., “ㅋㅋㅋㅋ”), numerals, and symbols—suggesting that **repeat/symbol normalization**, **byte fallback**, and **Unigram subword regularization** can close the gap.

V Conclusion

This work demonstrates that **tokenizer design** and **length policy** decisively influence performance of small RNN-based classifiers for Korean sentiment analysis. **MeCab (95p)** yields the best **Accuracy (0.8616)**, while **SentencePiece variants (BPE 8k/10k, Unigram 8k)** follow closely. Moving to **98p** degrades performance (**0.8418**)—consistent with increased padding and RNN sensitivity to longer sequences. For agglutinative Korean, respecting **morphological boundaries** shortens sequences and increases information density, which is particularly beneficial under SLM constraints (limited parameters/latency).

Future directions include: (a) **Transformer-tiny** comparisons under matched budgets, (b) quantifying padding-efficiency via **pack_padded_sequence**, (c) evaluating normalization/byte policies for **OOV/emoji/code-switching**, and (d) testing generalization across tasks and domains.

VI References

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